National Aeronautics and Space Administration

FY26 President and Director's Research and Development Fund (PDRDF)

A Growing Methylmercury Liability: Satellites Track a Decade of Artisanal Gold Mining Pond Growth in Madre de Dios, Peru

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2.0×
standing pond area in 2025 (13,061 ha) vs. the 2019–20 enforcement-era mean
2x the area of Pasadena

3.0×
pond-area growth over the decade (4,303 → 13,061 ha, 2016–2025)
1 ha = 100m² = 11x30m Landsat px

−44%
clearing on mining ground in the 2020 quarantine; recovered by July at record gold prices

6.1×
pond growth on the Madre de Dios floodplain, upstream of Puerto Maldonado since 2019

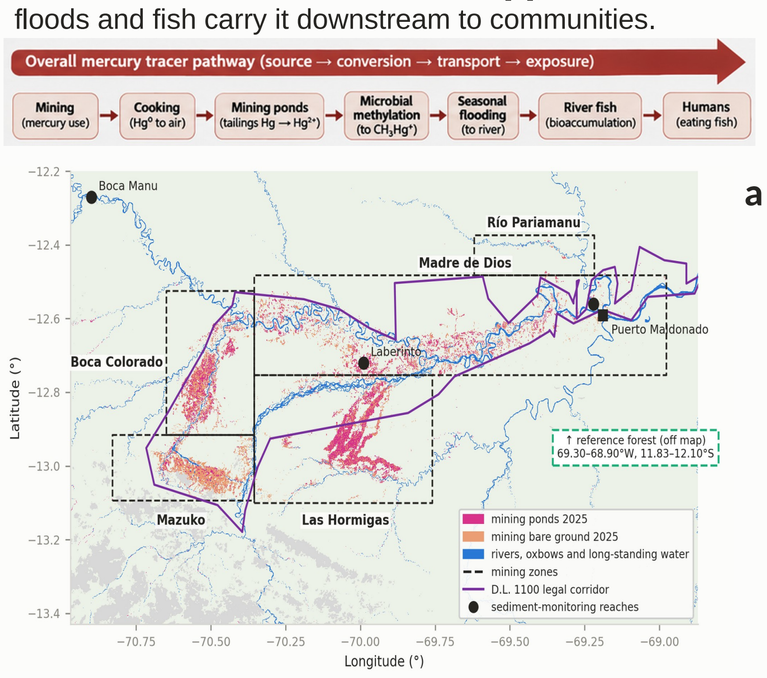
≈35,900 ha
of methylating river surface equivalent added since 2019–20

Objectives

- Measure from orbit how artisanal gold mining in Madre de Dios, Peru has grown since 2016, through 2019 Operation Mercurio mining crackdown and through 2020 COVID-19 pandemic as gold prices rose to record highs.
- Track **standing mining ponds** (and their toxic methyl-mercury production capacity), the primary variable that controls mercury risk — not just forest loss.

Background

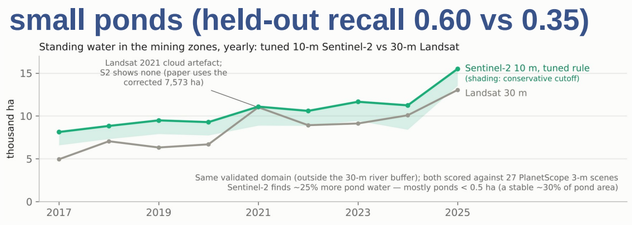
Artisanal gold mining is the largest human source of mercury [1]. In Madre de Dios, a biodiversity hotspot [3], **mining pits fill into stagnant ponds where bacteria convert spilled mercury to toxic methylmercury at 5.5× the rate of turbulent river water** [2]; seasonal floods and fish carry it downstream to communities.



Study region (~42,000 km²), five mining zones (dashed); 2025 ponds magenta, mining bare ground orange, legal corridor (D.L. 1100) purple; dots: river reaches of the suspended-sediment proxy.

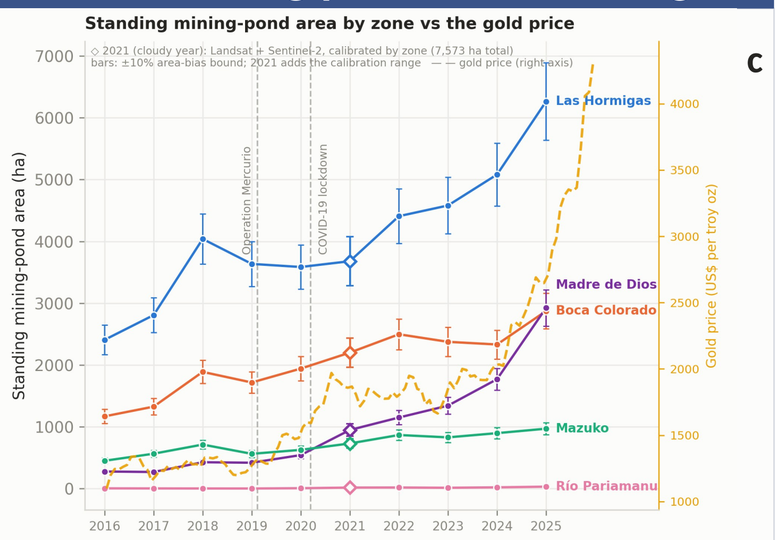
Approach

- **Annual 30-m Landsat** pond and bare-ground maps, 2016–2025, rivers, oxbows removed (RF, 91%)
- **Validated** on 585 blind labels (98% precision); **98% of new ponds** lie in Amazon Mining Watch scars [5]
- **3-m PlanetScope check:** Landsat ~15% low (1.15×, no trend); misses only ponds < 0.5 ha
- **New: tuned 10-m Sentinel-2 rule beats Landsat on small ponds (held-out recall 0.60 vs 0.35)**

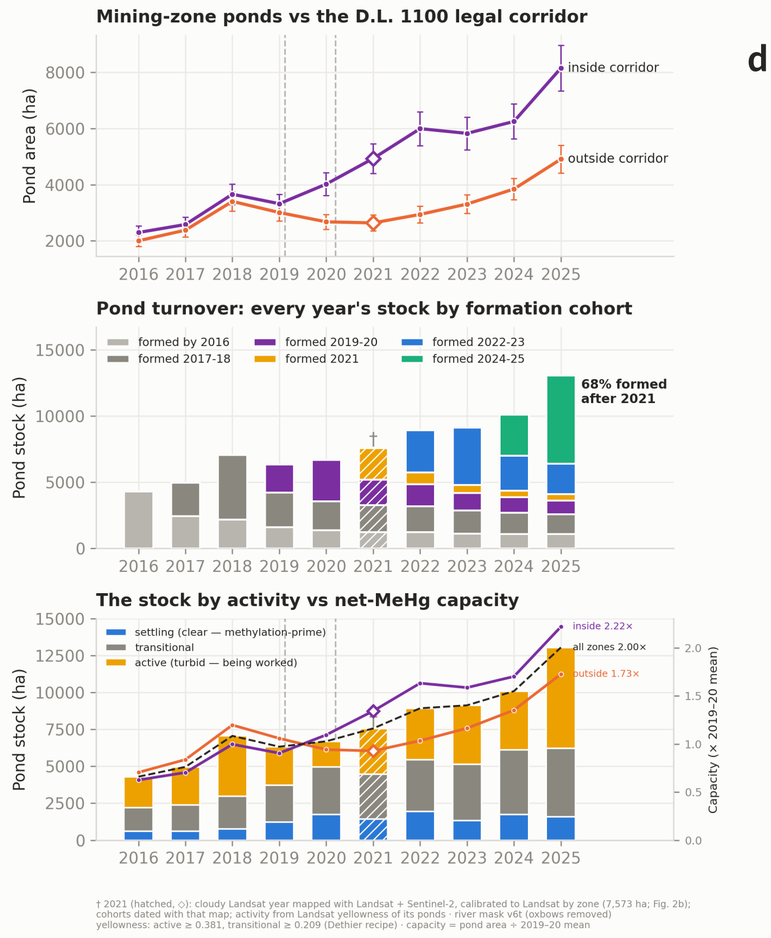


Tuned 10-m Sentinel-2 yearly series vs Landsat, same domain; both checked against 27 PlanetScope 3-m scenes.

Results: Mining ponds scale with gold



Gold more than tripled; pond area dipped only 10% under Operation Mercurio, held through COVID-19, and reached ~13,100 ha by 2025 — 3.0× the 2016 pond-scape, 2.0× the enforcement baseline.

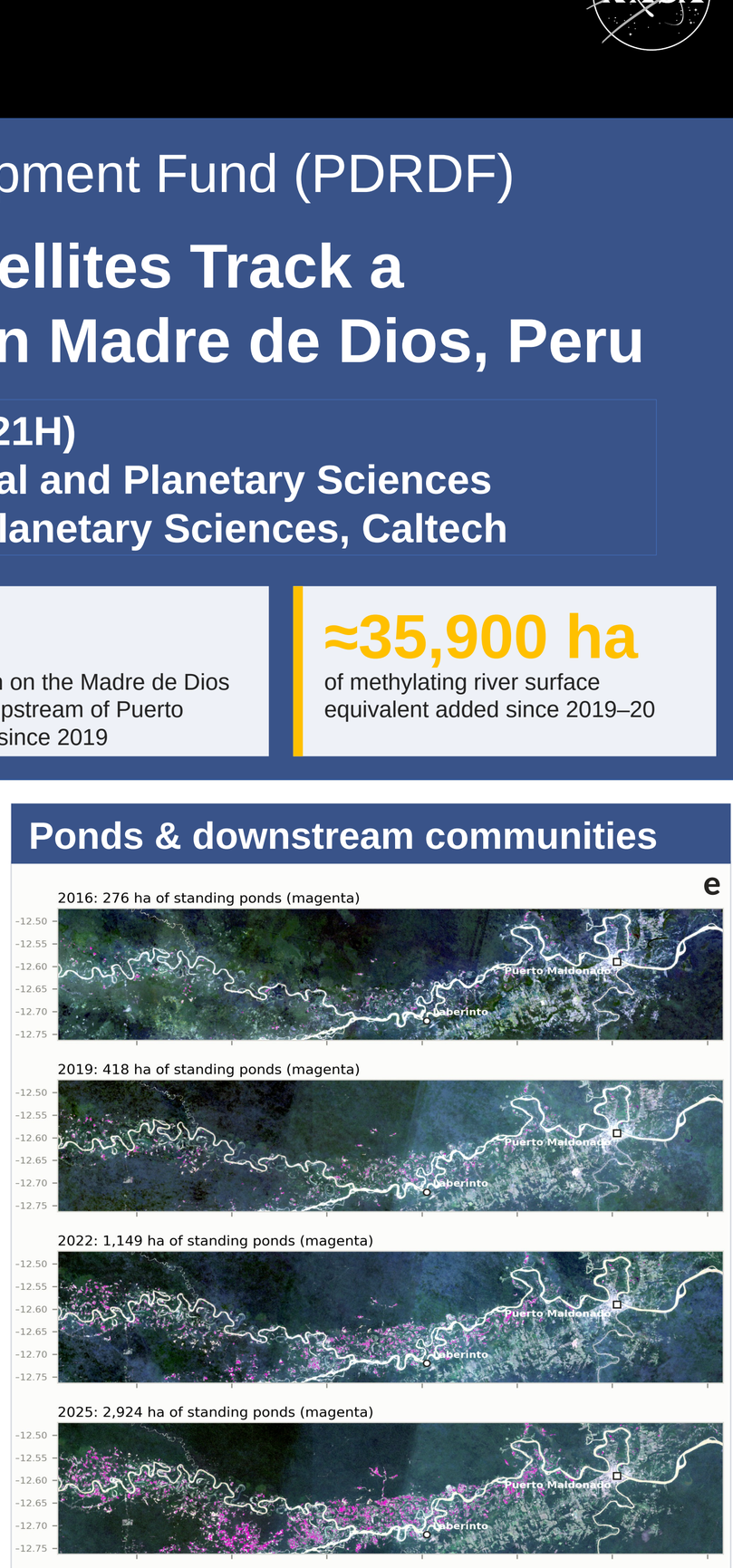


Enforcement cut actively worked (turbid) ponds 57% in 2018–2020 [4] but did not drain them: methylation-prime water nearly tripled (~2,200 → 6,200 ha); two-thirds of 2025 ponds formed after 2021.

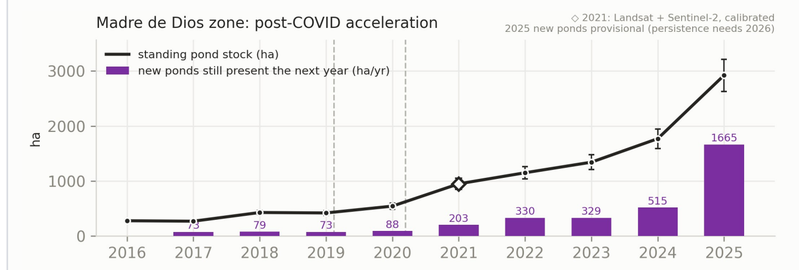
Significance to JPL/Caltech/NASA

- **Free Landsat, PlanetScope and Sentinel-1/2 can monitor artisanal mining's mercury hazard across a decade**
- A path to operational pond products for NASA Earth Action health and water applications, NISAR and SBG. Monitoring must track standing water, as stagnant pond liability increases even when clearing stops.
- **Next:** resample fish and hair mercury downstream; measure methylation in ponds of known age.

Ponds & downstream communities



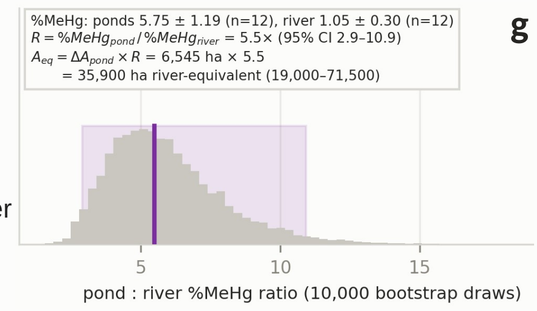
Floodplain ponds (magenta): 276 ha (2016) → 418 (2019 crackdown) → 2,924 (2025). Sentinel-2 imagery, Copernicus.

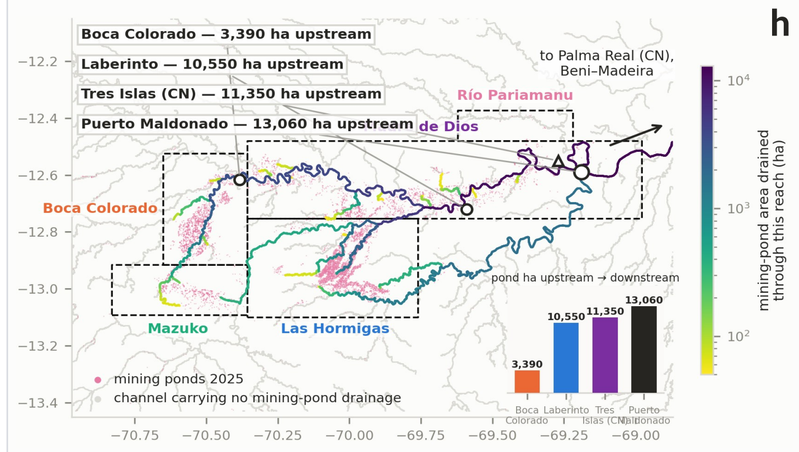


Methylating pond over river network

All 13,061 ha of 2025 ponds drain past Puerto Maldonado toward communities downstream.

Ponds hold 5.5× river %MeHg: the +6,545 ha since 2019–20 ≈ **35,900 ha** of river equivalent mercury.





Workflow

1 · DATA SOURCES

RAW IMAGERY (THIS STUDY)
- Landsat satellites, 30 m — NASA/USGS; 10 dry seasons, 2016–2025
- PlanetScope 3-m imagery — NASA

DERIVED BY OTHER GROUPS
- Radar alerts (RADD) · Nighttime lights (VIIRS)
- River network (MERIT) · Mining scars (AMW)

RECORDS
- Gold price · published fish-mercury data

classify →

2 · SATELLITE MAPPING
- Annual maps of water + bare ground spectral water test; cloud, shadow and terrain screening; most clear views agree
- Machine-learning river removal + a screen for natural river sandbars
- Pond yellowness tells activity (muddy vs clear)
- Ponds = standing water − rivers

verify →

3 · VALIDATION
- 2,245 hand-labeled checkpoints 585 water · 1,066 river/pond · 594 bare
- 3-m imagery check: Landsat ~15% low, no trend, so the totals are conservative
- 97% of new ponds fall inside independently mapped Amazon Mining Watch
- Cloudy 2021 filled with Sentinel-2 (7,573 ha)

aggregate →

4 · DERIVED PRODUCTS (THIS STUDY)
- Pond area and pond age, by zone and year, 2016–2025
- Worked / clearing / settled pond split
- Inside vs outside the legal corridor
- Monthly clearing + night-light series
- Pond area routed downriver to each community

interpret →

5 · FINDINGS
- Pond area doubled since the 2020 and tripled since 2016 crackdown — 13,061 ha by 2025
- COVID + enforcement cut activity, but the standing ponds never drained (clearing −44%, worked ponds −57%)
- Mercury capacity ≈ 35,900 ha of river
- Exposure ranked for 4 communities

National Aeronautics and Space Administration
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PRD-PDRDF-012
Clearance Number: CL#26-4012

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Publication: Anderson, N., Aguilar, I., Podest, E, Tejada, J. A growing methylmercury liability: standing mining ponds in Madre de Dios, 2016–2025. *In preparation (2026).*

Key references: [1] UNEP, Global Mercury Assessment 2018 (2019). [2] Gerson et al., Sci. Adv. 6, eabd4953 (2020). [3] Weinhouse et al., J. Expo. Sci. Environ. Epidemiol. 31, 126 (2021). [4] Dethier et al., Conserv. Lett. 16, e12978 (2023). [5] Earth Genome, Amazon Mining Watch (2026).

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